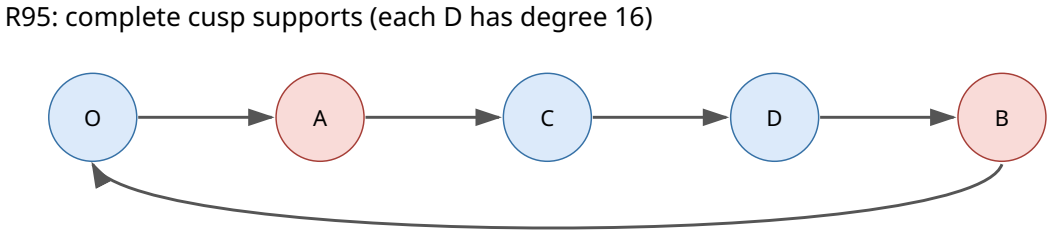
R95 cusp divisors for the two-chart cover

From the accepted degree-16 ramification fibers and the elliptic base divisor $\operatorname{div}(u) = [A]+[B]-2[O]$, the transported overlap functions have complete source divisors

$$\operatorname{div}(h) = D_O + D_D - 2D_B, \qquad \operatorname{div}(k) = D_O + D_C - 2D_A.$$

Here each D_X has degree 16 and includes the two index-1 and two index-7 branches above X. Both divisors have zero and pole degree 32.



On $U_0 = X \setminus (D_A \cup D_B)$, h is regular and vanishes exactly at $D_O \cup D_D$. Hence $U_0 \cap U_1 = D_{U_0}(h)$ and $U_0 \cup U_1 = X$ for $U_1 = X \setminus (D_O \cup D_D)$. The abstract second chart has overlap $D(k)$. R91/R94 check the transition on its rank-32 algebra. The Lean file checks the finite cusp permutation and support arithmetic; divisor pullback and the normal-chart identification use the cited geometric/source inputs.